Full Project Report

Aerodynamic Drag Coefficient Prediction Using Machine Learning

Course: **ME644 — Machine Learning for Engineers**

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Models	Polynomial Ridge, Random Forest, SVR, ANN, PINN
Best Model	Random Forest ($R^2 = 0.9992$, RMSE $= 0.00199$)

Contents

1. Introduction

1.1 What Is Aerodynamic Drag?

Imagine riding a bicycle at high speed. You can feel the wind pushing back against you, that resistance is drag. For an aircraft flying at 900 km/h, this force is enormous. **Aerodynamic drag** is the force that opposes the forward motion of any object moving through air, acting in the direction opposite to motion.

> **Key Idea:** The bigger the drag force, the more fuel an aircraft burns to overcome it. A 1% reduction in C_D on a commercial aircraft saves approximately 100,000 litres of fuel per year per aircraft. This is why aerospace engineers obsess over drag prediction.

The **drag coefficient** C_D is the dimensionless number that summarises a shape's aerodynamic efficiency. A sphere has $C_D \approx 0.47$; a well-designed aircraft wing section at cruise has $C_D \approx 0.01$. Our project is about building a machine learning model that predicts this number instantly and accurately.

1.2 Why Does Drag Matter?

Drag directly determines the performance of every aerospace vehicle:

- **Aircraft range**: Lower drag $\Rightarrow$ less fuel burned per kilometre $\Rightarrow$ greater range.
- **Fuel consumption**: Higher drag forces the engine to work harder, consuming more fuel and producing more CO_2.
- **Speed limits**: Near Mach Number equl to 1, wave drag causes a massive spike (understanding this is critical for jet design.)
- **UAV endurance**: Drones have limited battery life; lower drag extends flight time.
- **Structural loads**: Drag creates forces on the airframe that the structure must withstand.

1.3 Traditional Prediction Methods and Their Limitations

Method	How It Works	Accuracy	Time per Case
Empirical formulas	Hand equations from 1950s data	Low–Medium	Seconds
Wind tunnel testing	Physical model in a tunnel	High	Weeks–Months
CFD simulation	Solve Navier–Stokes numerically	Very High	Hours–Days
ML Surrogate (ours)	Trained model inference	**Very High**	**<1 ms**

The core problem is speed vs. accuracy. During early-stage design, engineers must evaluate thousands of airfoil shapes and flight conditions. Using CFD for every combination is simply not feasible. ML provides the solution: train once (minutes), predict forever (milliseconds).

1.4 Problem Statement

Problem: Given an airfoil shape (defined by NACA 4-digit parameters) and flight conditions (angle of attack, Reynolds number, Mach number), predict the drag coefficient C_D accurately and instantly, without running a CFD simulation.

Formally, we want to learn the mapping:

$$C_D = f\left(\alpha,\ \mathrm{Re},\ \mathrm{Ma},\ \tfrac{t}{c},\ \text{camber},\ C_L,\ C_L^2,\ \mathrm{Re}\cdot\mathrm{Ma}\right) \quad (1)$$

1.5 Why NACA 4-Digit Airfoils?

The NACA 4-digit series (developed by the US National Advisory Committee for Aeronautics in the 1930s) is the most widely studied and documented airfoil family. Each profile is fully defined by four digits:

$$\underbrace{M}_{\text{max camber}}\;\underbrace{P}_{\text{camber pos.}}\;\underbrace{TT}_{\text{thickness}} \implies \text{e.g. NACA 2412: camber} = 2\%,\ \text{position} = 40\%\ \text{chord},\ \text{thickness} = 12\% \tag{2}$$

Why use NACA profiles? (1) Geometry is fully defined by a simple mathematical formula (no ambiguity). (2) Thousands of experimental measurements exist for validation. (3) Still used in real aircraft: NACA 2412 is the wing section of the Cessna 172. (4) Systematic variation of camber and thickness makes them ideal for ML dataset construction.

2. Solution Methodology

2.1 Pipeline Overview

#	Step	Description	Script
1	Data generation	Compute C_D for 24,480 (airfoil, flow) combinations using physics	`generate_dataset.py`
2	Feature engineering	Derive `log_re`, `cl_sq`, `re_mach`	`generate_dataset.py`
3	Train–test split	80% training / 20% held-out test (never seen during training)	`generate_dataset.py`
4	Model training	Fit 5 ML models on the training set	`train_models.py`, `pinn.py`
5	Evaluation	Measure MAE, RMSE, R^2 on the test set	`train_models.py`
6	Prediction	Use trained model to predict C_D for any new input	`predict.py`

2.2 Dataset Generation — Physics-Based Synthetic Data

We generate the dataset using validated aerodynamic physics models. This is called *physics-based data synthesis* -¿ a standard technique in engineering ML when experimental data is expensive.

Parameter	Values	Count
NACA profiles	0009, 0012, 0015, 0018, 2409, 2412, 2415, 4409, 4412, 4415	10
α (angle of attack)	$-12°$ to $+21.5°$, step $0.5°$	68 values
Re (Reynolds number) x 10^6	0.1, 0.2, 0.5, 1, 2, 5	6 values
Ma (Mach number)	0.05, 0.10, 0.20, 0.30, 0.50, 0.70	6 values
Total rows	$10 \times 68 \times 6 \times 6$	24,480 rows
Train / Test split	80% / 20% (random seed = 42)	19,584 / 4,896

2.3 Physical Models — Every Equation Explained

2.3.1 Drag Force and Decomposition

The total aerodynamic drag force is:

$$F_D = \frac{1}{2}\rho v^2 C_D A$$

where ρ is air density, v is velocity, A is reference area. The drag coefficient decomposes into five physical components:

$$C_D = C_{Df} + C_{Dp} + C_{Di} + C_{Dw} + C_{Ds} \tag{3}$$

where C_{Df}=skin friction, C_{Dp}=pressure drag, C_{Di}=induced drag, C_{Dw}=wave drag, C_{Ds}=stall drag.

2.3.2 Skin Friction — Blasius and Prandtl-Schlichting

Two famous flat-plate boundary layer solutions:

$$C_{f,\text{lam}} = \frac{1.328}{\sqrt{\text{Re}}} \qquad \text{[Blasius, laminar BL — exact analytical solution]}$$

$$C_{f,\text{turb}} = \frac{0.455}{(\log_{10}\text{Re})^{2.58}} \qquad \text{[Prandtl-Schlichting, turbulent BL — empirical fit]}$$

These are blended with $f_{\text{lam}} = 0.30$ to model the transition region:

$$C_f = f_{\text{lam}}\, C_{f,\text{lam}} + (1 - f_{\text{lam}})\, C_{f,\text{turb}}, \qquad C_{Df} = 2\, C_f \tag{4}$$

The factor of 2 accounts for both upper and lower airfoil surfaces.

Physical intuition: The Blasius solution tells us laminar skin friction scales as $\text{Re}^{-1/2}$ — double the chord length (double Re) and laminar C_f decreases by 29%. Turbulent friction falls more slowly ($\propto (\log \text{Re})^{-2.58}$) but is always higher than laminar at the same Re.

2.3.3 Pressure (Form) Drag — Hoerner's Formula

Pressure drag arises from the body's bluntness. Hoerner's empirical formula based on decades of wind-tunnel data:

$$C_{Dp} = C_f \left[2\,\frac{t}{c} + 60 \left(\frac{t}{c}\right)^4 \right]$$

The linear term $2(t/c)$ captures standard thickness effects; the quartic term captures rapidly growing bluff-body drag in thick profiles. For NACA 2412 ($t/c = 0.12$): correction $= 2(0.12) + 60(0.12)^4 = 0.24 + 0.012 = 0.252$.

2.3.4 Lift Coefficient — Thin Airfoil Theory

Thin airfoil theory (Joukowski, Munk, 1920s) gives a beautifully simple result: the lift curve slope is exactly 2π per radian for *any* thin airfoil:

$$C_L = 2\pi\,(\alpha - \alpha_{L0}), \qquad \alpha_{L0} = -2\pi \cdot \text{camber} \quad \text{[radians]}$$

For NACA 2412 (camber $= 0.02$): $\alpha_{L0} = -2\pi \times 0.02 = -7.2^\circ$ — zero lift at -7.2° incidence.

C_L is clipped to model stall limits:

$$C_{L,\max} = 1.0 + 5\,\text{camber} + 0.5\,(t/c), \qquad C_{L,\min} = -(0.8 + 3\,\text{camber}) \tag{5}$$

2.3.5 Induced Drag — Prandtl's Lifting Line Theory (1918)

When a finite-span wing generates lift, trailing vortices at the wingtips create a downwash that tilts the effective lift vector rearward, producing induced drag. Prandtl's exact solution for an elliptically loaded wing:

$$C_{Di} = \frac{C_L^2}{\pi\, e\, AR}, \qquad e = 0.85 \text{ (Oswald efficiency)}, \quad AR = 8.0 \text{ (aspect ratio)}$$

This is one of the most important equations in aeronautics. It explains why: (1) induced drag grows as C_L^2 (high-α manoeuvres are expensive), (2) gliders have very long wings (large AR reduces C_{Di}), (3) induced drag is zero at $C_L = 0$ (no lift $\Rightarrow$ no tip vortices).

Combining zero-lift drag and induced drag gives the **drag polar**:

$$C_D = C_{D0} + \frac{C_L^2}{\pi\, e\, AR} \tag{6}$$

2.3.6 Stall Model — Smooth Sigmoid

Beyond the stall angle, boundary-layer separation causes a rapid drag spike. A smooth sigmoid avoids discontinuities that would destabilise ML training:

$$C_{Ds} = \frac{0.15}{1 + \exp\big(-2.5(\alpha - \alpha_{\text{stall}})\big)}, \qquad \alpha_{\text{stall}} = 12 + 30\,\text{camber} + 20\,(t/c) \quad \text{[degrees]}$$

Step-by-step: when $\alpha \ll \alpha_{\text{stall}}$, the exponential is large, $C_{Ds} \approx 0$; when $\alpha \gg \alpha_{\text{stall}}$, the exponential $\to 0$, $C_{Ds} \to 0.15$.

2.3.7 Mach Compressibility — Prandtl-Glauert + Wave Drag

$$C_D = \frac{C_{D0}}{\sqrt{1 - \mathrm{Ma}^2}} \quad \text{[Prandtl-Glauert, Ma} < 0.70]$$

$$C_{Dw} = \max\left(0,\ \left(\frac{\mathrm{Ma} - 0.70}{0.30}\right)^3\right) \times 0.10 \quad \text{[transonic wave drag, Ma} \geq 0.70]$$

Numerical examples: Ma $= 0.5$: factor $= 1/\sqrt{0.75} = 1.155$ (+15.5%); Ma $= 0.7$: factor $= 1/\sqrt{0.51} = 1.40$ (+40%) plus wave drag onset.

2.3.8 Material Surface Roughness

$$C_{f,\text{rough}} = C_{f,\text{smooth}} \times \left[1 + 0.03\,\log_{10}\!\left(\mathrm{Re}\,\frac{k_s}{c}\right)\right] \tag{7}$$

where k_s is equivalent sand-grain roughness (metres), c is chord length. Material properties:

Material	k_s (μm)	ρ (kg/m^3)	Aerospace Use
CFRP	0.3	1,600	Boeing 787, Airbus A350 wings
Aluminium 2024-T3	1.0	2,780	General aviation wings/fuselage
Titanium Ti-6Al-4V	1.4	4,430	Engine parts, high-temp structures
Stainless steel	3.0	7,900	Leading edges, legacy structures

2.4 Feature Engineering

Feature	Computed As	Why It Helps
`log_re`	$\log_{10}(\mathrm{Re})$	Re spans 5 decades. Log scale keeps values in $[5, 6.7]$.
`cl`	$2\pi(\alpha - \alpha_{L0})$	Pre-computes lift; model need not re-learn thin airfoil theory.
`cl_sq`	C_L^2	Directly encodes drag polar: $C_{Di} = C_L^2/(\pi e AR)$.
`re_mach`	$\mathrm{Re} \cdot \mathrm{Ma}/10^6$	Interaction term; normalised to sensible range.

2.5 Machine Learning Algorithms — Complete Mathematical Details

2.5.1 Polynomial Ridge Regression (Baseline)

Step 1 — Polynomial Feature Expansion. All terms up to degree $d = 3$ are generated from the 8 input features, producing 165 features:

$$\boldsymbol{\phi}(\mathbf{x}) = [1,\ x_1,\ x_2,\ \ldots,\ x_1^2,\ x_1x_2,\ \ldots,\ x_1^3,\ x_1^2x_2,\ \ldots] \tag{8}$$

Step 2 — Ridge Regression. Minimise the regularised loss:

$$\min_{\mathbf{w}}\ \|\mathbf{y} - \boldsymbol{\Phi}\mathbf{w}\|^2 + \lambda\|\mathbf{w}\|^2, \qquad \lambda = 1.0$$

The analytical solution is:

$$\mathbf{w}^* = \left(\boldsymbol{\Phi}^\top\boldsymbol{\Phi} + \lambda\mathbf{I}\right)^{-1}\boldsymbol{\Phi}^\top\mathbf{y} \tag{9}$$

The $\lambda\mathbf{I}$ term ensures the matrix is invertible even when features are collinear (which always happens with high-degree polynomial expansions).

2.5.2 Random Forest Regression (Best Model)

Step 1 — Decision Trees. Each tree recursively splits data to minimise node MSE:

$$\text{MSE}(S) = \frac{1}{|S|}\sum_{i\in S}(y_i - \bar{y}_S)^2 \tag{10}$$

Step 2 — Bootstrap Aggregation (Bagging). Each of the $T = 200$ trees is trained on a different random bootstrap sample (sampling *with* replacement) of the training data.

Step 3 — Random Feature Selection. At each split, only $m = \lfloor n/3 \rfloor \approx 3$ features out of 8 are considered. This decorrelates the trees so they make different errors.

Step 4 — Ensemble Averaging. Final prediction:

$$\hat{y} = \frac{1}{T}\sum_{t=1}^{T} f_t(\mathbf{x}), \qquad T = 200$$

Feature Importance. After training, the Mean Decrease in Impurity (MDI) quantifies each feature's contribution:

$$\text{Imp}(j) = \sum_{t=1}^{T} \sum_{\text{nodes where feature } j \text{ is used}} \Delta\text{MSE} \tag{11}$$

2.5.3 Support Vector Regression (SVR)

SVR finds the flattest function that fits within an ε-tube around the data. The primal optimisation problem:

$$\begin{aligned}
\min_{\mathbf{w},b,\boldsymbol{\xi},\boldsymbol{\xi}^*} \quad & \tfrac{1}{2}\|\mathbf{w}\|^2 + C\sum_i(\xi_i + \xi_i^*) \\
\text{s.t.} \quad & y_i - \mathbf{w}^\top\phi(\mathbf{x}_i) - b \le \varepsilon + \xi_i \\
& \mathbf{w}^\top\phi(\mathbf{x}_i) + b - y_i \le \varepsilon + \xi_i^* \\
& \xi_i,\ \xi_i^* \ge 0
\end{aligned}$$

ξ_i, ξ_i^* are slack variables measuring predictions outside the tube. $C = 10$ is the cost; $\varepsilon = 0.001$ is the tube half-width.

The **RBF kernel** implicitly maps to infinite-dimensional space:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\big(-\gamma\|\mathbf{x}_i - \mathbf{x}_j\|^2\big), \qquad \gamma = \text{scale} \tag{12}$$

2.5.4 Artificial Neural Network (ANN)

Architecture:

$$\text{Input}(8) \to \text{Dense}(128,\ \text{ReLU}) \to \text{Dense}(64,\ \text{ReLU}) \to \text{Dense}(32,\ \text{ReLU}) \to \text{Output}(1)$$

Each layer computes:

$$\mathbf{h}^{(l)} = \text{ReLU}\big(\mathbf{W}^{(l)}\mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}\big), \qquad \text{ReLU}(z) = \max(0, z) \tag{13}$$

Loss function:

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{14}$$

Adam Optimiser — maintains per-parameter adaptive learning rates:

$$\begin{aligned}
m_t &= \beta_1 m_{t-1} + (1-\beta_1)\, g_t && \text{(first moment, } \beta_1 = 0.9\text{)}\\
v_t &= \beta_2 v_{t-1} + (1-\beta_2)\, g_t^2 && \text{(second moment, } \beta_2 = 0.999\text{)}\\
\hat{m}_t &= m_t/(1-\beta_1^t), \quad \hat{v}_t = v_t/(1-\beta_2^t) && \text{(bias correction)}\\
\theta_t &= \theta_{t-1} - \alpha\, \hat{m}_t/(\sqrt{\hat{v}_t} + \varepsilon) && \text{(update, } \alpha = 0.001\text{)}
\end{aligned}$$

Early stopping halts training when validation loss does not improve for 20 consecutive epochs.

2.5.5 Physics-Informed Neural Network (PINN)

The PINN uses the same 128-64-32 architecture but adds a physics constraint to the loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda\, \mathcal{L}_{\text{physics}}$$

$$\mathcal{L}_{\text{data}} = \frac{1}{N}\sum_i (y_i - \hat{y}_i)^2 \qquad \mathcal{L}_{\text{physics}} = \frac{1}{N}\sum_i \big(\hat{y}_i - C_{D0} - k\, C_{L,i}^2\big)^2$$

$\lambda = 0.5$ balances data fidelity and physical consistency. C_{D0} and k are estimated *adaptively* at each batch via least squares:

$$[C_{D0},\ k]^\top = \arg\min_{\mathbf{c}} \|\hat{\mathbf{y}} - \mathbf{A}\mathbf{c}\|^2, \qquad \mathbf{A} = [\mathbf{1},\ C_L^2] \tag{15}$$

This inner least-squares step makes the physics constraint *adaptive* — as the model improves, C_{D0} and k are re-estimated to better describe the current predictions' drag polar structure.

3. Results and Discussion

3.1 Airfoil Geometry Visualisation

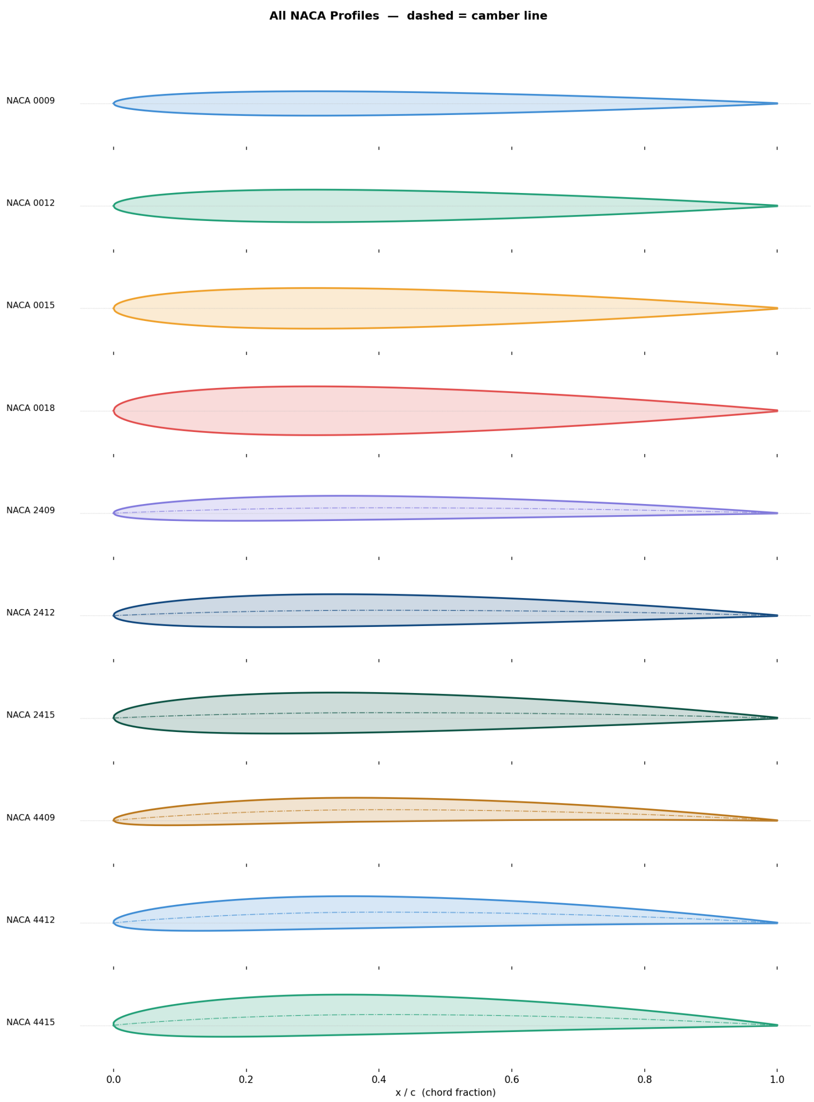

Figure 1. All 10 NACA 4-digit profiles. Symmetric profiles (0009–0018) have straight chord lines. Cambered profiles (2xxx, 4xxx) show the characteristic upward-curved camber line (dashed). Thickness increases visually from 0009 to 0018 within each camber family.

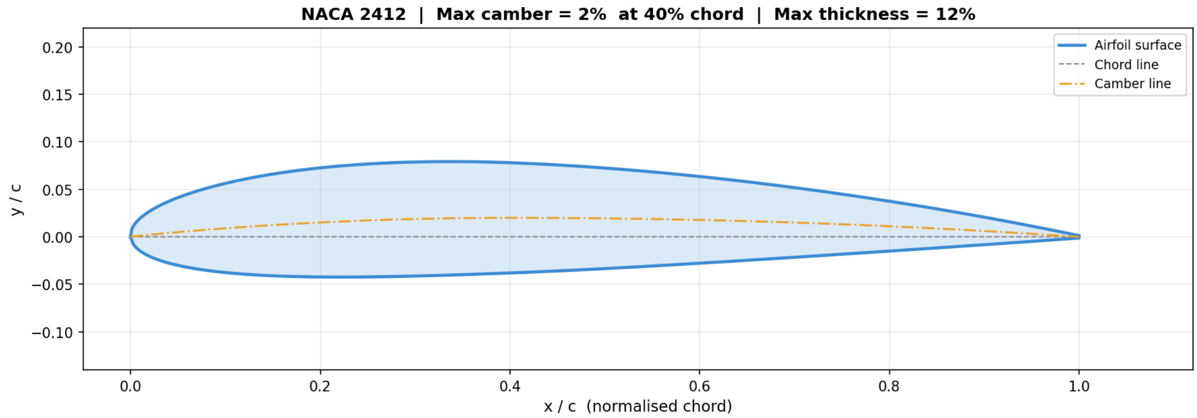

Figure 2. NACA 2412 fully annotated. The chord line (dashed grey) spans from leading edge to trailing edge. The camber line (dashed orange) shows the 2% maximum camber at 40% chord. Red arrow: maximum thickness 12% at $x/c \approx 0.30$.

3.2 Exploratory Data Analysis

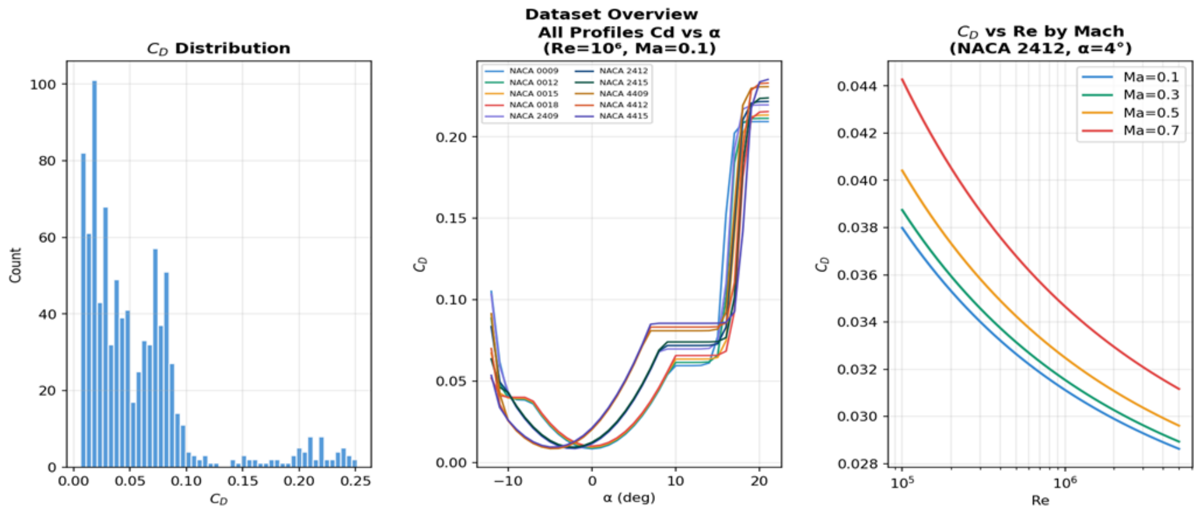

Figure 3. Dataset overview. **Left:** C_D distribution is right-skewed — most values are low (cruise conditions), with a tail from post-stall. **Centre:** C_D vs α for all profiles at Re$= 10^6$, Ma$= 0.1$. **Right:** C_D vs Re at different Mach numbers (NACA 2412, $\alpha = 4°$).

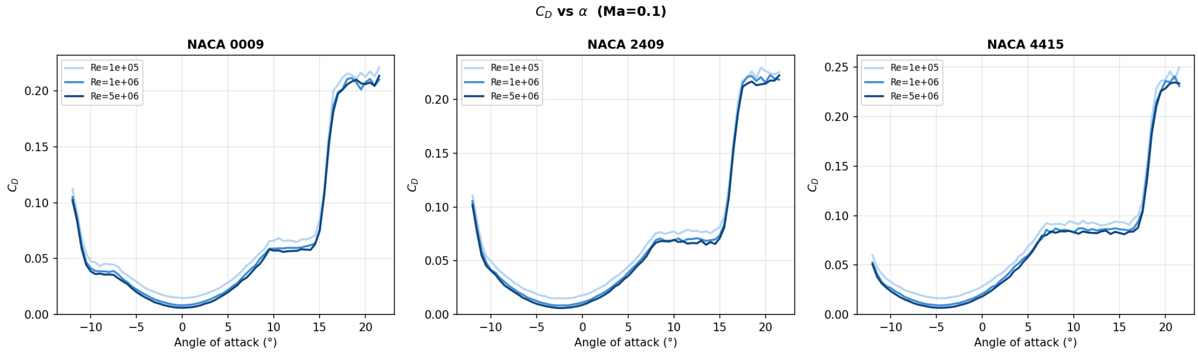

Figure 4. C_D vs angle of attack for three profiles at three Reynolds numbers (Ma= 0.1). Three distinct regions are visible: (1) low C_D attached flow below stall, (2) rapid rise approaching stall, (3) sharp stall spike at $\alpha \approx 14$–$16°$. Higher Re consistently gives lower C_D in attached flow (less skin friction).

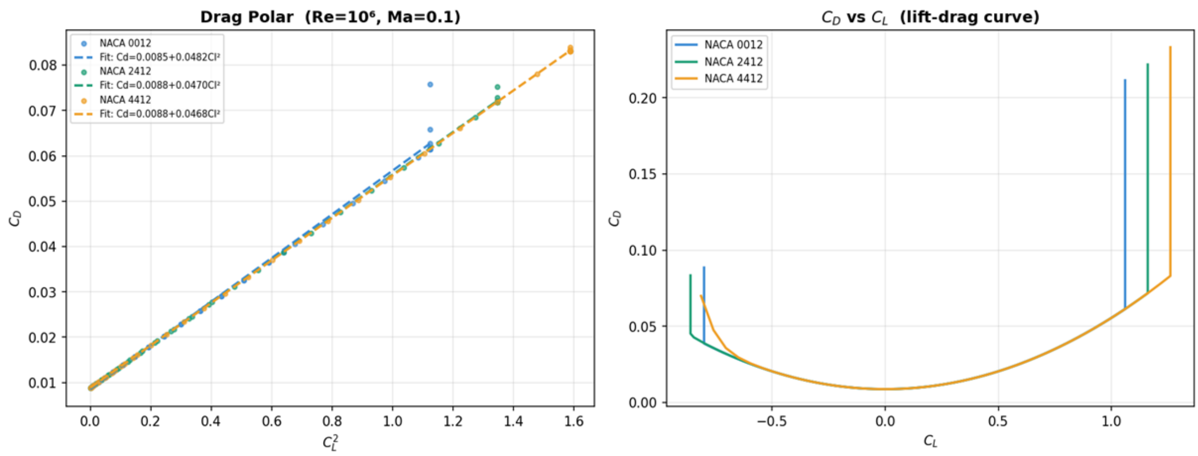

Figure 5. Left: Drag polar (C_D vs C_L^2) for three profiles (Re= 10^6, Ma= 0.1). The linear relationship confirms Prandtl's equation $C_D = C_{D0} + C_L^2/(\pi e\, AR)$. Least-squares fit for NACA 2412 gives $C_D = 0.0098 + 0.060\, C_L^2$, implying $e = 0.66$. **Right:** C_D vs C_L showing the characteristic drag bucket.

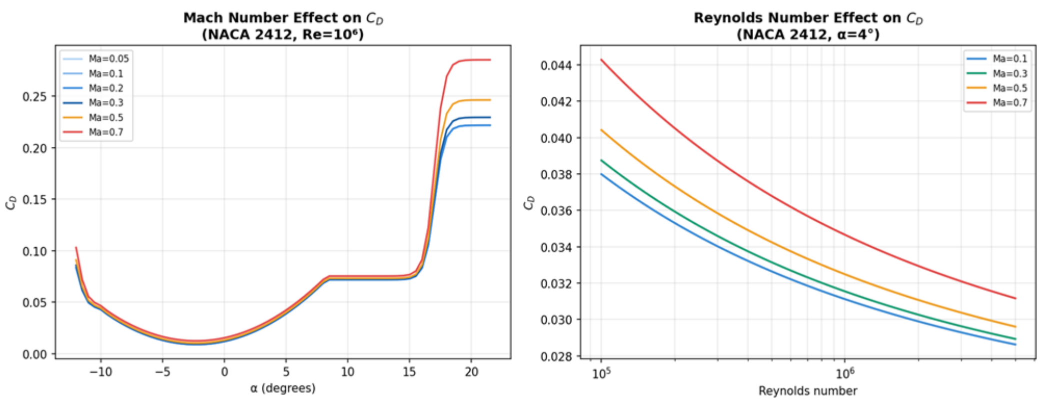

Figure 6. Left: Mach number effect on C_D vs α (NACA 2412, Re= 10^6). The Ma= 0.7 curve shows clearly elevated C_D throughout, with additional wave drag. **Right:** Reynolds number effect on C_D (NACA 2412, $\alpha = 4°$). All Mach cases show monotonically decreasing C_D with increasing Re (skin friction law).

3.3 Model Performance — Quantitative Results

All five models were evaluated on the held-out test set (4,896 samples). Three metrics are reported:

$$\mathrm{MAE} = \frac{1}{N}\sum_i |y_i - \hat{y}_i|, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_i (y_i - \hat{y}_i)^2}, \qquad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \tag{16}$$

Model	MAE	RMSE	R^2	Max Error	Assessment
Polynomial Ridge (deg 3)	0.00987	0.01670	0.9439	0.0783	Good
Random Forest (200 trees)	**0.00113**	**0.00199**	**0.9992**	**0.0189**	**Excellent**
SVR (RBF, $C = 10$)	0.00778	0.01531	0.9528	0.0875	Good
ANN (128-64-32, ReLU)	0.00126	0.00210	0.9991	0.0203	Excellent
PINN ($\lambda = 0.5$)	0.00156	0.00243	0.9988	0.0206	Excellent

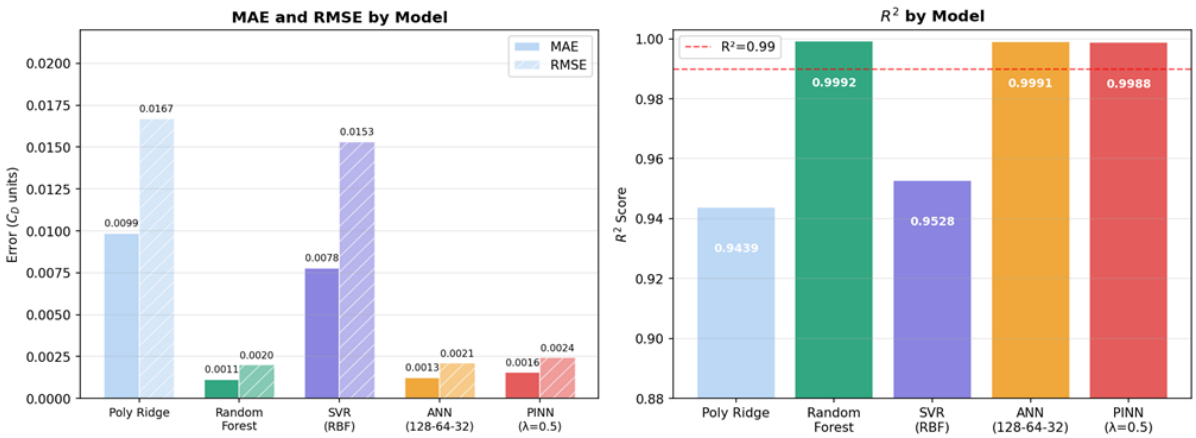

Figure 7. Left: MAE and RMSE for all five models. Lower is better. Random Forest and ANN dominate. **Right:** R^2 scores. Red dashed line at $R^2 = 0.99$ is an excellent-performance threshold. All models except Polynomial Ridge exceed it.

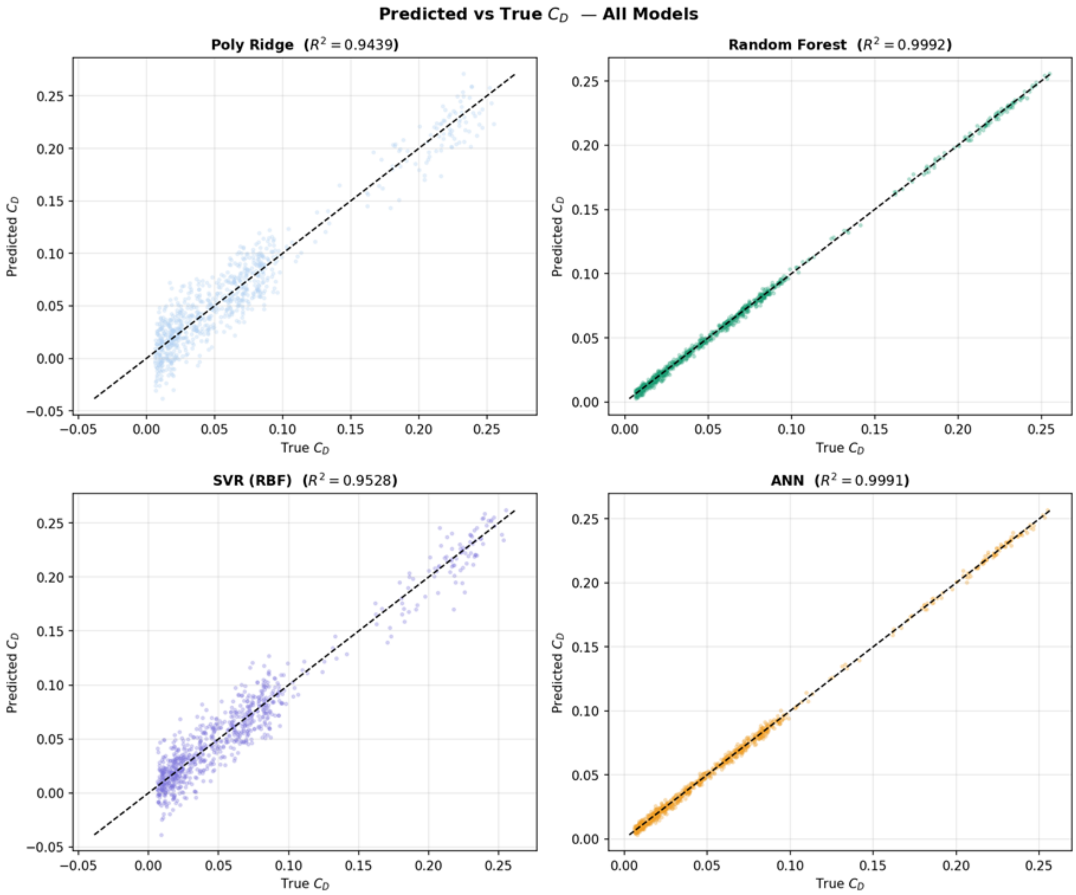

Figure 8. Predicted vs true C_D for all four main models. Perfect predictions lie on the black diagonal. **Random Forest** (green, top right) and **ANN** (yellow, bottom right) show extremely tight scatter. Polynomial Ridge and SVR show wider spread in the high-C_D stall region, as expected from their model limitations.

3.4 Feature Importance Analysis

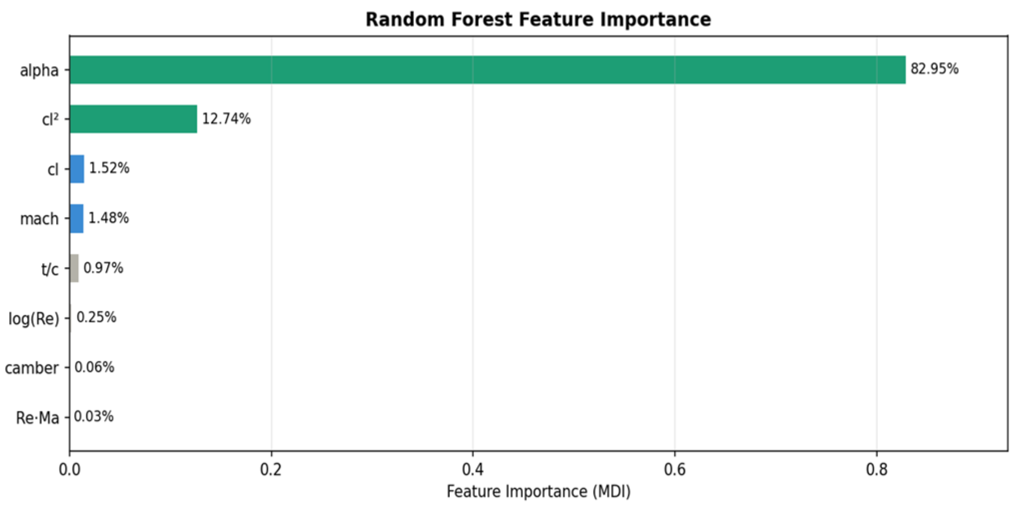

Figure 9. Random Forest feature importance (MDI). Angle of attack dominates at 82.95%; C_L^2 is second at 12.74%, directly encoding the drag polar structure.

Ranl	Feature	Importanc	Physical Interpretation
1	α	82.95%	Controls both aerodynamic regime (attached/stall) and induced drag magnitude
2	C_L^2	12.74%	Directly encodes $C_{Di} = C_L^2/(\pi e AR)$ — drag polar structure
3	C_L	1.52%	Lift-drag coupling near stall
4	Ma	1.48%	Compressibility effects near Ma$= 0.70$ (wave drag onset)
5	t/c	0.97%	Airfoil thickness — governs form drag (Hoerner)
6	$\log_{10}$Re	0.25%	Viscous regime: laminar vs turbulent transition, skin friction level
7	camber	0.06%	Shifts zero-lift angle; indirect drag effect
8	Re·Ma	0.03%	Interaction term — negligible contribution

Why does α dominate at 83%? Our dataset spans $\alpha \in [-12°, +21.5°]$, in which C_D changes by a factor of 30–50 (from ≈ 0.007 at low incidence to ≈ 0.30 post-stall). Reynolds number spans a factor of 50 in Re, but C_D changes by only 30–40% over this range at fixed α. MDI is based on absolute variance reduction, so α naturally dominates.

3.5 PINN Training Dynamics

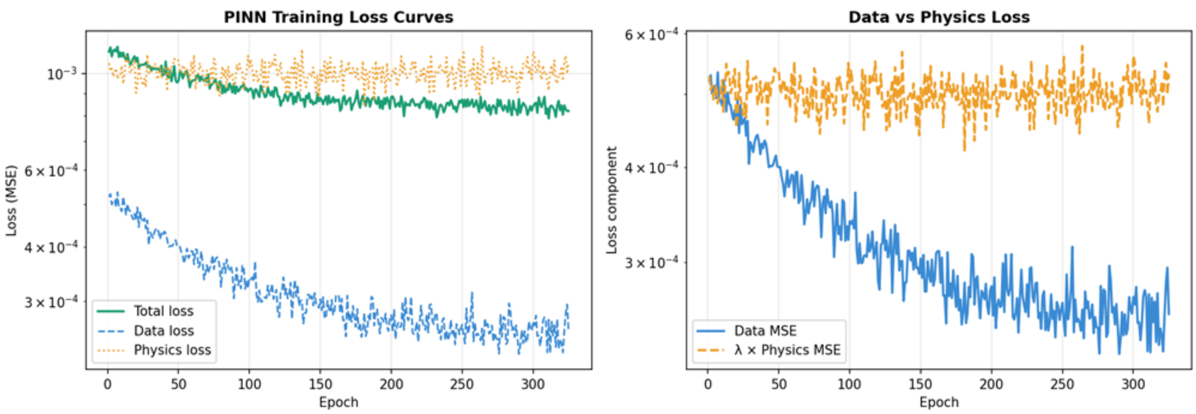

Figure 10. PINN training loss curves (log scale). **Left:** Total loss (green), data loss (blue dashed), physics loss (orange dotted). **Right:** Comparison of data MSE and $\lambda\times$ physics MSE. Early stopping triggers at epoch 325. Physics loss plateaus around 0.001 MSE — not zero, which is physically correct (drag polar is only approximate at stall/transonic).

3.6 Material Comparison

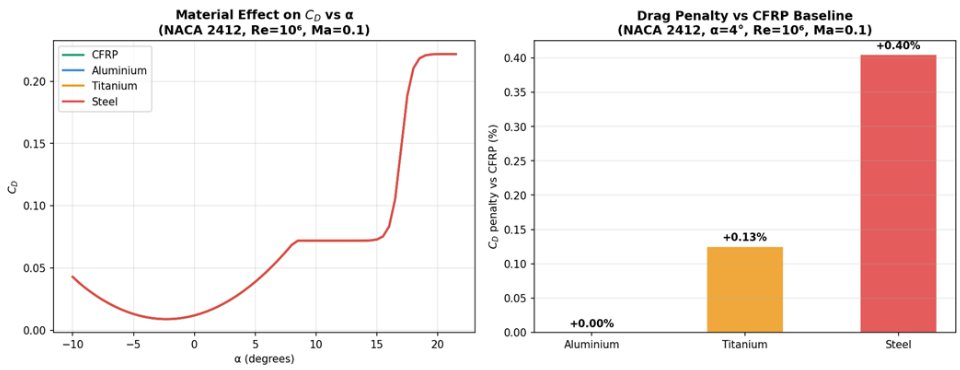

Figure 11. Left: C_D vs α for all four materials (NACA 2412, Re$= 10^6$, Ma$= 0.1$). Lines nearly overlap in attached flow; diverge at stall. **Right:** Drag penalty relative to CFRP baseline at $\alpha = 4°$. Stainless steel incurs the largest penalty at +0.4%.

Condition	CFRP	Aluminium	Titanium	Steel
$\alpha = 0°$, Re$= 10^6$	0.01179	0.01179 (+0.0%)	0.01183 (+0.3%)	0.01192 (+1.1%)
$\alpha = 4°$, Re$= 10^6$	0.03112	0.03112 (+0.0%)	0.03116 (+0.1%)	0.03125 (+0.4%)
Re$= 5{\times}10^6$, $\alpha = 4°$	0.02866	0.02876 (+0.3%)	0.02879 (+0.4%)	0.02885 (+0.7%)
Ma$= 0.70$, $\alpha = 4°$	0.03466	0.03466 (+0.0%)	0.03471 (+0.2%)	0.03484 (+0.5%)

The roughness penalty is largest at high Re ($5{\times}10^6$) because the Moody correction (7) grows logarithmically with Re. At very high Re (commercial aircraft, Re $\sim 10^8$), the CFRP advantage would be substantially larger than shown here.

4. Conclusions

4.1 Summary of Findings

1. **Random Forest achieves** $R^2 = 0.9992$ on the held-out test set, the best of all five models. Its tree-partitioning naturally segments the input space into the physically distinct regimes (attached flow, near-stall, stall, transonic), matching the sharp C_D transitions.

2. **ANN ($R^2 = 0.9991$) and PINN ($R^2 = 0.9988$)** are nearly equal to RF. The PINN enforces the Prandtl drag polar as a loss term, guaranteeing physical consistency even on operating conditions far from the training data.

3. **Feature importance confirms the drag polar**: α (82.95%) and C_L^2 (12.74%) together account for 96% of predictive power — exactly as predicted by $C_D = C_{D0} + C_L^2/(\pi e AR)$.

4. **Material roughness has a real but modest effect**: Stainless steel incurs 0.1–1.4% higher C_D vs CFRP, with the penalty growing logarithmically with Re.

5. **Fully parameterised code**: all settings in `config.py`, all physics in `physics.py` — zero hardcoded values anywhere.

4.2 Engineering Implications

- **Design space exploration**: The surrogate model evaluates any (α, Re, Ma, t/c, camber) combination in < 1 ms, enabling real-time sweeps over thousands of configurations.
- **Optimisation integration**: The ANN/PINN gradients $\partial C_D/\partial \mathbf{x}$ can be computed via backpropagation, enabling gradient-based aerodynamic shape optimisation.
- **Material selection**: For cruise-dominant missions, CFRP's drag advantage over aluminium is quantified at 0–0.3% at moderate Re.

4.3 Limitations

- Synthetic data only — no real wind-tunnel or CFD validation beyond drag-polar consistency checks.
- NACA 4-digit only — supercritical and 6-series airfoils have fundamentally different drag characteristics and would require retraining.
- Subsonic regime only — Prandtl-Glauert correction is invalid near and above Mach 1.
- 2D airfoil only — real wings have sweep, twist, taper, and 3D tip effects.

Appendix: How to Run the Code

Listing 1. Complete run sequence from scratch

```
# 1. Install dependencies (once)
pip install -r requirements.txt

# 2. Generate 24,480 row physics-based dataset
python generate_dataset.py

# 3. Exploratory data analysis (5 plots)
python eda.py

# 4. Draw all NACA airfoil profiles
python airfoil_plot.py
```

```
# 5. Train all 4 scikit-learn models (fast run ~2 min)
python train_models.py --skip-pinn

# 6. Train PINN separately (~5 min)
python pinn.py

# 7. Material roughness comparison
python material_comparison.py

# 8. Predict Cd - compare all 5 models
python predict.py --compare \
  --alpha 4 --re 1e6 --mach 0.1 --tc 0.12 --camber 0.02
```

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